IMO 2026

Complete Solutions

DeepSeek V4 Pro

Independently graded: **19/42**

Autonomous run, no human intervention.

Solutions as written by the model; all six problems.

`github.com/deedy/imo-2026`

IMO 2026 — Problem 1

Solution by DeepSeek V4 Pro

Source: `results/deepseek-v4-pro/problem-01/current.md`. Rendered as verbatim source (the current.md file is authoritative).

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# imo-2026-01 — tracking file

## Status

solved

## Problem

There are $2026$ integers greater than $1$ written on a blackboard, not necessarily different. In a move, Confucius
chooses two integers $m>1$ and $n>1$ from different places on the blackboard and replaces these two integers with
$\{}gcd(m,n)$ and $\{}frac{\{}mathrm{lcm}(m,n)}{\{}gcd(m,n)}$. He continues to make moves while it is possible to
do so.

(a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer $M$ on the
blackboard is greater than $1$.

(b) Prove that the value of $M$ does not depend on the choices of Confucius.

## Approaches tried

- Brute-force simulation for small cases to guess the final value $M$ and invariants.

- Discovered that for each prime $p$, the gcd of the exponents of $p$ across all numbers is invariant, and the final
$M$ equals $\{}prod_p p^{\{}gcd(\{}text{exponents})}$.

- Termination proved via monovariant $K+\{}sum\{}Omega$ strictly decreasing (where $K$ counts numbers $>1$, $\{}Omega$
counts prime factors with multiplicity).

- Invariant proof uses the fact that on the exponent multiset the move acts as $(\{}alpha,\{}beta)\{}mapsto(\{}min(\{}alpha,\{
which preserves the gcd of the whole multiset.

## Current best

The final unique integer $M>1$ equals

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M = \{}prod_{p\{}text{ prime}} p^{\{},d_p},\{}qquad

d_p = \{}gcd\{}bigl(v_p(a_1),v_p(a_2),\{}dots,v_p(a_{2026})\{}bigr),

\{}]

where $a_1,\{}dots,a_{2026}$ are the initial numbers and $v_p(a)$ is the exponent of $p$ in $a$. Both parts are
proved below.

## Full proof

Let $N=2026$ and let the initial numbers be $a_1,a_2,\{}dots,a_N$ (all integers $>1$). At any moment we denote the
numbers on the board by $x_1,\{}dots,x_N$ (some may become $1$ during the process). The order of the numbers is
irrelevant, so we may treat the board as a multiset.

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### 1. Termination — proof of part (a)

For a positive integer $t$ let $\{}Omega(t)$ be the number of prime factors of $t$ counted with multiplicity; set
$\{}Omega(1)=0$. Define

\{}[

K = \{}#\{}{\{},i \{}mid x_i > 1\{},\{}},\{}qquad

S = \{}sum_{i=1}^{N} \{}Omega(x_i).

\{}]

Both $K$ and $S$ are non-negative integers.

A move picks two distinct indices $i,j$ with $x_i,x_j>1$. Put $m=x_i$, $n=x_j$. Let

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g=\{}gcd(m,n),\{}qquad h=\{}frac{\{}operatorname{lcm}(m,n)}{\{}gcd(m,n)}=\{}frac{mn}{g^{2}}.

\{}]

The two numbers m,n are replaced by g and h. Write $m=g a$, $n=g b$ with $\{}gcd(a,b)=1$; then $h=ab$.

We analyse how K and S change.

* **Case 1: $g=1$.**

Then $h=mn>1$. The new pair is $(1,\{},mn)$. Consequently

\{}[

K' = K-1,\{}qquad

S' = S-\{}Omega(m)-\{}Omega(n)+\{}Omega(1)+\{}Omega(mn)=S.

\{}]

(Equality holds because $\{}gcd(m,n)=1$ implies $\{}Omega(mn)=\{}Omega(m)+\{}Omega(n)$.)

* **Case 2: $g>1$ and $m=n$.**

Then $a=b=1$, so $h=1$. The new pair is $(g,1)$. Hence

\{}[

K' = K-1,\{}qquad

S' = S-2\{}Omega(m)+\{}Omega(g)+\{}Omega(1)=S-\{}Omega(m)\{}le S-1.

\{}]

(Here $\{}Omega(m)=\{}Omega(g)$ because $m=g$.)

* **Case 3: $g>1$ and $m\{}neq n$.**

Then a,b are not both 1, so $h=ab>1$. Both $g>1$ and $h>1$ hold.

\{}[

K' = K,\{}qquad

S' = S-\{}Omega(m)-\{}Omega(n)+\{}Omega(g)+\{}Omega(h).

\{}]

Since $\{}Omega(m)=\{}Omega(g)+\{}Omega(a)$, $\{}Omega(n)=\{}Omega(g)+\{}Omega(b)$ and $\{}Omega(h)=\{}Omega(ab)=\{}Omega(a)+\
(using $\{}gcd(a,b)=1$), we obtain

\{}[

S' = S - \{}Omega(g) \{}le S-1.

\{}]

In every case either K decreases by 1 (Cases 1, 2) or K stays the same and S decreases by at least 1 (Case 3). Therefore the quantity

\{}[

\{}Phi = K+S

\{}]

strictly decreases by at least 1 with each move. Being a non-negative integer, $\{}Phi$ cannot decrease forever; after finitely many moves the process must stop.

The process stops exactly when fewer than two numbers on the board exceed 1, i.e. $K\{}le 1$. We claim that K can never become 0. Indeed, a move can produce at most one 1 (Cases 1 and 2 yield exactly one 1; Case 3 yields none), so K decreases by at most 1 per move. Initially $K=N\{}ge 2$. To reach $K=0$ we would need at least N moves that reduce K, but after $N-1$ such moves we already have $K=1$; at that point no further move is possible because a move requires two numbers >1. Hence the process necessarily terminates with $K=1$. Thus after finitely many moves exactly one integer $M>1$ remains on the board. This proves part (a).

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2. Invariance of the exponent-gcd - proof of part (b)

Fix an arbitrary prime p. For an integer $t\{}ge 1$ denote by $v_p(t)$ the exponent of p in the prime factorisation of t; set $v_p(1)=0$.

Consider a move on numbers m,n with $\{}alpha=v_p(m)$, $\{}beta=v_p(n)$. Then

\{}[

v_p(\{}gcd(m,n)) = \{}min(\{}alpha,\{}beta),\{}qquad

v_p\{}!\{}left(\{}frac{\{}operatorname{lcm}(m,n)}{\{}gcd(m,n)}\{}right)

= \{}alpha+\{}beta-2\{}min(\{}alpha,\{}beta)=|\{}alpha-\{}beta|.

\{}]

Thus for the multiset $\{}mathcal E_p$ of the N values $v_p(x_1),\{}dots,v_p(x_N)$, a move replaces two elements $\{}alpha,\{}beta$ by $\{}min(\{}alpha,\{}beta)$ and $|\{}alpha-\{}beta|$, leaving all other elements unchanged.

For a multiset S of non-negative integers, let $\{}gcd(S)$ denote the greatest common divisor of its elements, defined as the unique non-negative integer d such that $d\{}mid s$ for every $s\{}in S$ and every common divisor of S divides d. (If all elements are 0 then $\{}gcd(S)=0$.)

Lemma (invariance). Let S be a multiset of non-negative integers and let S' be obtained from S by picking two elements $\{}alpha,\{}beta$ and replacing them with $\{}min(\{}alpha,\{}beta)$ and $|\{}alpha-\{}beta|$. Then $\{}gcd(S')=\{}gcd(S)$.

Proof. Put $d=\{}gcd(S)$. From $d\{}mid\{}alpha$ and $d\{}mid\{}beta$ we get $d\{}mid\{}min(\{}alpha,\{}beta)$ and $d\{}mid|\{}alpha-\{}beta|$; together with d dividing every other element of S unchanged, we obtain $d\{}mid\{}gcd(S')

Conversely, let $d'=\{}gcd(S')$. Then $d'\{}mid\{}min(\{}alpha,\{}beta)$ and $d'\{}mid|\{}alpha-\{}beta|$. If $\{}alpha\{}le\{ then $\{}alpha=\{}min(\{}alpha,\{}beta)$ and $\{}beta=\{}min(\{}alpha,\{}beta)+|\{}alpha-\{}beta|$, so $d'\{}mid\{}alpha$ and $d'\{}mid\{}beta$. The case $\{}beta\{}le\{}alpha$ is symmetric. Hence d' divides all elements of S, whence $d'\{}mid d$.

Thus $d'=d$, as claimed. □

Apply this lemma to the multiset $\{}mathcal E_p$. Initially

\{}[

\{}mathcal E_p^{(0)} = \{}{v_p(a_1),\{},v_p(a_2),\{},\{}dots,\{},v_p(a_N)\{}},

\{}qquad

d_p := \{}gcd\{}bigl(\{}mathcal E_p^{(0)}\{}bigr).

\{}]

By the lemma, d_p is invariant throughout the whole process.

When the process ends, part (a) tells us that the board contains exactly one number $M>1$ together with $N-1$ ones. Hence

\{}[

\{}mathcal E_p^{\{}text{(final)}} = \{}{\{},v_p(M),\{},\{}underbrace{0,\{},0,\{},\{}dots,\{},0}_{N-1\{}text{ times}}\{},\{}}.

\{}]

The gcd of a multiset consisting of a number v and several zeros is v itself (because $\{}gcd(v,0,\{}dots,0)=v$; every divisor of v divides 0, and the largest positive divisor of v is v itself). Therefore

\{}[

v_p(M) = \{}gcd\{}bigl(\{}mathcal E_p^{\{}text{(final)}}\{}bigr) = d_p \{}qquad\{}text{for every prime } p.

\{}]

Consequently

\{}[

M = \{}prod_{p\{}text{ prime}} p^{\{},d_p},

\{}qquad\{}text{where}\{}qquad

d_p = \{}gcd\{}bigl(v_p(a_1),v_p(a_2),\{}dots,v_p(a_N)\{}bigr).

\{}]

The right-hand side depends only on the initial N integers $a_1,\{}dots,a_N$, not on the particular sequence of moves chosen by Confucius. This completes part (b).

Remark. The number 2026 plays no special role; the same statements hold for any $N\{}ge 2$ initial integers greater than 1.

IMO 2026 — Problem 2

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

partial

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle $\angle LBA$, L lies inside the angle $\angle ACK$, and

$$\angle KBA = \angle ACL, \qquad \angle LBK = \angle LNC, \qquad \angle LCK = \angle BMK.$$

Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Approaches tried

- Numeric search + gradient descent: solutions exist and satisfy OM=ON to within numerical error (~1e-7). Earlier wrong circumcenter function gave false non-zero; corrected function confirms OM=ON.
- Sympy polynomial algebra: derived the condition OM=ON is equivalent to numerator P = 0, where P is a polynomial of degree 3 in coordinates (x1,y1,x2,y2,a1,a2). Trying to prove P is in the ideal generated by the three angle conditions.

- Special case (right isosceles A=(0,0), B=(2,0), C=(0,2)): solved exactly. Found two solution branches: one has x1=y2 (which forces OM=ON), the other also forces OM=ON after substituting interior conditions.
- Attempted to find geometric meaning: spiral similarities centered at O or at A don't match. The three conditions might correspond to an isogonal conjugate or a special point.

Current best

Fixed coordinate setup: M=0, N=1, B = -A, C = 2 - A. Then: - OM = ON $2 \cdot \mathrm{Re}(O) = 1$ P = 0 where P = numerator of (2X-1). - P = a1^2y1 - a1^2y2 - a1y1 + a1y2 + a2^2y1 - a2^2y2 - a2x1^2 + a2x1 + a2x2^2 - a2x2 - a2y1^2 + a2y2^2 + x1^2y2 - x1y2 - x2^2y1 + x2y1 + y1^2y2 - y1y2^2

The three angle conditions are (with U=K+A, V=L+A): I: Im(A(A-1) / (U(V-2))) = 0 II: Im(U(L-1) / (V(1-A))) = 0 III: Im(A(U-2) / (K(V-2))) = 0

These give three polynomial equations R1=R2=R3=0.

Need to show P Ideal(R1,R2,R3) or at least that P=0 when the interior conditions hold.

Full proof

(TODO)

IMO 2026 — Problem 3

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n, determine the largest value c such that Liu Bang may guarantee a total length of at least c, regardless of Xiang Yu's play.

Solution document

Status

partial

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n, determine the largest value c such that Liu Bang may guarantee a total length of at least c, regardless of Xiang Yu's play.

Approaches tried

- Brute-force discrete search for small n: n=1 gives 2/3; n=2 gives 3/5=0.6; n=3 appears around 4/7 0.5714 but discrete search suggests possibly lower.
- Hypothesized formula: c = (n+1)/(2n+1). For n=1: 2/3 correct. For n=2: 3/5 correct. For n=3: 4/7 seems plausible but needs verification.

- Attempted to compute X's optimal response to L's marks via discretized search; results suggest L can guarantee at least (n+1)/(2n+1) by placing marks at 1/(2n+1), 2/(2n+1), ..., n/(2n+1).

Current best

Conjecture: Liu Bang can guarantee c = (n+1)/(2n+1). The strategy is to mark points at 1/(2n+1), 2/(2n+1), ..., n/(2n+1). Xiang Yu's optimal response seems to be to also mark n points at the remaining positions n+1/(2n+1), ..., 2n/(2n+1), making all pieces equal to 1/(2n+1). Then Liu Bang gets n+1 pieces out of 2n+1, totaling (n+1)/(2n+1). Need to prove both that Liu Bang can guarantee at least this much (lower bound) and that Xiang Yu can prevent Liu Bang from getting more (upper bound).

We need a rigorous proof: - Upper bound: Xiang Yu can limit Liu Bang to at most (n+1)/(2n+1). Strategy for X? - Lower bound: Liu Bang's strategy ensures at least (n+1)/(2n+1).

Full proof

(To be completed)

IMO 2026 — Problem 4

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

partial

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Approaches tried

- Initial analysis of one-move win conditions. Realized Mulan needs both resulting triangles to be winning.

- Quick discretized search for some values (depth 3-4, limited split points). Results:
  - 45°, 60°, 90° appear winning (no counterexample found).
  - 30° mostly winning but (91,44,45) loses.
  - 36° appears winning.
  - 72°, 108°, 120°, 150° lose from many states.
- Formalized the operation algebraically: split angle a into p, q; triangle options are (p, b, c+q) and (q, c, b+p).
- One-move win analysis: Mulan can win in one move if she can split such that both resulting triangles contain .
  - This is possible when = 90° (if both other angles < 90°, split as 90 - b and 90 - c).
  - Also possible if some angle = k ? But need both triangles to have .
- Search too slow to explore thoroughly. Need a theoretical characterization.

Current best

We suspect the answer might be: Mulan can force a win if and only if is not a divisor of 180° in the sense that 180°/ is not an integer? Or perhaps = 90° is the only winning? But search suggests 36°, 45°, 60° also winning. Let's compute more precisely.

The state space is continuous; discretization may miss moves. Need a mathematical invariant or strategy.

Full proof

(Not yet complete)

IMO 2026 — Problem 5

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that

$$\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$$

for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

1. Numerical brute-force — only $f(x) = x + c$ ($c \geq 0$) survives.
2. Substituting $x = f(y)$ forces the functional equation $f(f(x)) = 2f(x) - x$.
3. Reformulating with $g(x) = f(x) - x$ yields two algebraic identities for the squared-inequality remainders A, B.
4. From $|A - B| \leq A + B$ we obtain a universal inequality (7).
5. The functional equation implies forward orbits are arithmetic progressions with common difference $g(x)$. If $g(x) < 0$ the orbit leaves $\mathbb{R}_{>0}$ – contradiction. Hence $g \geq 0$.
6. Using arithmetic progressions together with (7) shows that g attains at most one positive value.
7. A backward-induction lemma gives a point $u \in (0, c]$ with $g(u) = c$ (if $c > 0$ is a value).

8. A covering argument (intervals around the forward orbit) forces the zero-set $S_0 = \emptyset$, hence g is constant c.
9. Verification that $f(x) = x + c$ works for any $c \geq 0$.

Current best

Answer: $f(x) = x + c$ for a constant $c \geq 0$.

Full proof

Let $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ satisfy

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)} \qquad \forall x, y > 0. \tag{1}$$

All quantities are positive, so squaring preserves the inequalities:

$$2x^2 + 2f(y)^2 \geq (f(x) + y)^2 \geq 4xf(y) \qquad \forall x, y > 0. \tag{2}$$

---

1. A functional equation

Put $x = f(y)$ (which is > 0) in (2). The leftmost term becomes $2f(y)^2 + 2f(y)^2 = 4f(y)^2$ and the rightmost becomes $4f(y)f(y) = 4f(y)^2$. Hence the chain collapses to equality:

$$\big(f(f(y)) + y\big)^2 = 4f(y)^2.$$

Taking square roots (all terms are positive) yields

$$f(f(y)) + y = 2f(y) \qquad \Longrightarrow \qquad \boxed{f(f(x)) = 2f(x) - x \qquad \forall x > 0.} \tag{3}$$

Define

$$g(x) = f(x) - x \qquad (x > 0).$$

Then $g(x) > -x$ and (3) rewrites as

$$x + g(x) + g(x + g(x)) = 2x + 2g(x) - x \implies \boxed{g(x + g(x)) = g(x) \qquad \forall x > 0.} \tag{4}$$

---

2. Positivity of g

Lemma 0. $g(x) \geq 0$ for all $x > 0$.

Proof. Fix $x > 0$ and define the forward orbit $x_0 = x$, $x_{n+1} = f(x_n)$ for $n \geq 0$. Using (3),

$$x_{n+2} = f(x_{n+1}) = 2f(x_n) - x_n = 2x_{n+1} - x_n.$$

The recurrence $x_{n+2} - 2x_{n+1} + x_n = 0$ has the unique solution $x_n = A + Bn$ with A, B determined by the initial conditions: $x_0 = x$, $x_1 = f(x) = x + g(x)$. Hence $A = x$, $B = g(x)$, and

$$x_n = x + ng(x) \qquad (n \geq 0).$$

From (4) we have $g(x_n) = g(x)$ for all $n \geq 0$. Now, if $g(x) < 0$, then x_n becomes non-positive for sufficiently large n, contradicting that every x_n is in the domain $\mathbb{R}_{>0}$ (because $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$). Therefore $g(x) \geq 0$.

Thus g takes only non-negative values.

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3. Algebraic reformulation

Write $a = g(x)$, $b = g(y)$; then $f(x) = x + a$, $f(y) = y + b$. The two inequalities in (2) are

$$\begin{aligned} A(x, y) &:= 2x^2 + 2(y + b)^2 - (x + y + a)^2 \geq 0, \\ B(x, y) &:= (x + y + a)^2 - 4x(y + b) \geq 0. \end{aligned}$$

Expanding:

$$\begin{aligned} A(x, y) &= (x - y)^2 - a^2 - 2a(x + y) + 4by + 2b^2, \\ B(x, y) &= (x - y)^2 + a^2 + 2a(x + y) - 4bx. \end{aligned}$$

Adding and subtracting gives two fundamental identities

$$A(x, y) + B(x, y) = 2\big(x - y - b\big)^2 = 2\big(x - f(y)\big)^2, \qquad (5)$$
$$A(x, y) - B(x, y) = -2(a - b)(a + b + 2x + 2y). \qquad (6)$$

Because $A, B \geq 0$, we have $|A - B| \leq A + B$ (the absolute difference of two non-negative numbers never exceeds their sum). Substituting (5) and (6) and cancelling a factor 2 yields

$$\boxed{|g(x) - g(y)|\,\big(g(x) + g(y) + 2x + 2y\big) \leq \big(x - y - g(y)\big)^2 \qquad \forall x, y > 0.} \qquad (7)$$

Swapping x and y gives the symmetric inequality

$$|g(x) - g(y)|\,\big(g(x) + g(y) + 2x + 2y\big) \leq \big(y - x - g(x)\big)^2 \qquad \forall x, y > 0. \qquad (8)$$

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4. At most one positive value of g

Let $V = \{g(x) : x > 0\}$. Suppose V contains two distinct positive numbers $c_1 > c_2 > 0$. Choose $u, v > 0$ with $g(u) = c_1$, $g(v) = c_2$. From the functional equation (4) we obtain

$$g(u + nc_1) = c_1, \qquad g(v + mc_2) = c_2 \qquad \forall\, n, m \in \mathbb{N}_0.$$

Insert $x = u + nc_1$, $y = v + mc_2$ into (7). With $a = c_1$, $b = c_2$,

$$(c_1 - c_2)(c_1 + c_2 + 2u + 2v + 2nc_1 + 2mc_2) \le \left(u - v - c_2 + nc_1 - mc_2\right)^2. \qquad (9)$$

We show that for suitably chosen $n, m \to \infty$ the inequality fails.

Rational ratio. If $c_1/c_2 \in \mathbb{Q}$, write $c_1/c_2 = p/q$ with $p, q \in \mathbb{N}$ coprime. Then $pc_2 = qc_1$. Take $n = qt$, $m = pt$ for $t \in \mathbb{N}$. Then $nc_1 - mc_2 = 0$, so the right-hand side of (9) is the constant $(u - v - c_2)^2$, while the left-hand side grows linearly with t (coefficient $2(c_1 - c_2)(qc_1 + pc_2) > 0$). For large t the inequality is violated – contradiction.

Irrational ratio. If $c_1/c_2 \notin \mathbb{Q}$, the fractional parts $\{n(c_1/c_2)\}$ are dense in $[0, 1]$ (a classical fact: for irrational α, the sequence $n\alpha \bmod 1$ is dense). Consequently the set $\{nc_1 - mc_2 : n, m \in \mathbb{N}_0\}$ is dense in $\mathbb{R}$. Hence we can choose sequences $(n_k, m_k) \to (\infty, \infty)$ with $n_k c_1 - m_k c_2 \to -(u - v - c_2)$. Then the right-hand side of (9) tends to 0 while the left-hand side tends to $+\infty$, again a contradiction.

Thus V contains **at most one** positive number. Because $g \ge 0$, the only possible values are 0 and at most one $c > 0$. Consequently, either

- (i) $g \equiv 0$ (constant), which gives $f(x) = x$, a solution; or
- (ii) there is $c > 0$ such that $V \subseteq \{0, c\}$ with both values attainable.

We now exclude case (ii). Assume $c > 0$ and both 0 and c occur. Set

$$S_0 = \{x > 0 : g(x) = 0\}, \qquad S_c = \{x > 0 : g(x) = c\},$$

both non-empty. (Because $V \subseteq \{0, c\}$, every $x > 0$ belongs to $S_0 \cup S_c$.)

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5. Inequalities between the two level sets

Take $x \in S_0$, $y \in S_c$ ($a = 0$, $b = c$). The formulas for A and B become

$$\begin{aligned}
&A(x, y) = (x - y)^2 + 4cx + 2c^2 \ge 0 \quad \text{(always true)},\\
&B(x, y) = (x - y)^2 - 4cx \ge 0 \implies (x - y)^2 \ge 4cx, \qquad (10)\\
&A(y, x) = (x - y)^2 - c^2 - 2c(x + y) \ge 0 \implies (x - y)^2 \ge c^2 + 2c(x + y). \qquad (11)
\end{aligned}$$

From (11) and $x, y > 0$ we obtain $(x - y)^2 > c^2$, hence

$$|x-y|>c \qquad \forall\, x\in S_0,\ y\in S_c. \qquad (12)$$

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6. Existence of a small element in S_c

Lemma 1. There exists $u\in(0,c]$ with $g(u)=c$.

Proof. Pick any $y_0\in S_c$. If $y_0\le c$ we are done. Suppose $y_0>c$ and set $z=y_0-c>0$. Apply inequality (7) with $x=y_0$, $y=z$. We have $a=g(y_0)=c$, and $b=g(z)$ belongs to $\{0,c\}$ (all values are in $V\subseteq\{0,c\}$).

Compute:

$$\begin{aligned}\text{LHS}&=|c-b|(c+b+2y_0+2z)=|c-b|(c+b+2y_0+2(y_0-c))=|c-b|(b-c+4y_0),\\ \text{RHS}&=\big(y_0-z-b\big)^2=\big(y_0-(y_0-c)-b\big)^2=(c-b)^2.\end{aligned}$$

Let $d=b-c$. Then LHS $=|d|(d+4y_0)$, RHS $=d^2$.

- If $d>0$: $d(d+4y_0)\le d^2 \implies 4y_0d\le 0$, impossible.
- If $d=0$: inequality holds as $0\le 0$.
- If $d<0$: $|d|=-d$, giving $(-d)(d+4y_0)\le d^2 \implies -d^2-4y_0d\le d^2 \implies -4y_0d\le 2d^2$. Since $d<0$, divide by $-2d>0$: $2y_0\le -d=|d|$.

Thus $|d|\ge 2y_0$. But $b=g(z)\in\{0,c\}$, so $|d|=|b-c|$ is either 0 (if $b=c$) or c (if $b=0$).

If $b=0$, then $|d|=c$, giving $c\ge 2y_0$, i.e. $y_0\le c/2$, contradicting $y_0>c$. Therefore $b\ne 0$, so $b=c$; hence $g(z)=c$.

Thus $z=y_0-c\in S_c$. If $z>c$, repeat the argument with y_0 replaced by z. After $\lfloor y_0/c\rfloor$ steps we reach an element $u\in(0,c]$ belonging to S_c.

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7. The forward orbit and a covering argument

From Lemma 1 we have $u\in(0,c]$ with $g(u)=c$. By (4),

$$g(u+nc)=c \qquad \forall\, n\in\mathbb{N}_0.$$

Thus the arithmetic progression

$$P=\{u+nc : n\ge 0\}\subseteq S_c.$$

Consider the closed intervals

$$I_n=[\,u+nc-c,\ u+nc+c\,] \qquad (n\ge 0).$$

Their union covers all positive real numbers. Indeed, * $I_0 = [u - c,\ u + c]$; because $0 < u \le c$, the left endpoint is ≤ 0, so I_0 contains $(0,\ u + c]$. * $I_1 = [u,\ u + 2c]$, $I_2 = [u + c,\ u + 3c]$, ... Since I_n and I_{n+1} overlap (their intersection is $[u + nc,\ u + (n + 1)c]$), we have

$$\bigcup_{n\ge 0} I_n = (0, \infty).$$

Now take any $x > 0$. There exists $n \ge 0$ such that $x \in I_n$, i.e.

$$|x - (u + nc)| \le c.$$

If $x \in S_0$, then by (12) we must have $|x - (u + nc)| > c$ (because $u + nc \in S_c$). This is impossible. Hence $S_0 = \emptyset$.

---

8. Conclusion

Since $S_0 = \emptyset$, the function g never takes the value 0. Together with §4 this forces $g(x) \equiv c$ for the positive constant c. Thus

$$f(x) = x + c \qquad (c > 0).$$

The earlier case $g \equiv 0$ gives $f(x) = x$, which corresponds to $c = 0$. Both are covered by the single formula

$$\boxed{f(x) = x + c \quad \text{for some constant } c \ge 0.}$$

Verification. For $f(x) = x + c$ ($c \ge 0$) the original inequalities become

$$\sqrt{\frac{x^2 + (y + c)^2}{2}} \ge \frac{x + y + c}{2} \ge \sqrt{x(y + c)},$$

which are exactly the Quadratic-Mean–Arithmetic-Mean inequality and the Arithmetic-Mean–Geometric-Mean inequality applied to the two positive numbers x and $y + c$. Both inequalities are true for all $x, y > 0$.

Therefore the set of all solutions is $\{x \mapsto x + c \mid c \ge 0\}$.

IMO 2026 — Problem 6

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y.)

Solution document

Status

partial — developing rigorous proof

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n.

Approaches tried

- Computational exploration: sequences appear to be eventually periodic with constant difference; observed two regimes (arithmetic progression when a single prime dominates, or periodic sequence with period >1 governed by a finite "core" set of primes).
- Minimal hitting set / transversal hypergraph approach: each term's prime set must intersect all previous prime sets; the sequence is exactly the sorted list of numbers whose prime set contains one of the minimal transversals of the infinite family.
- The set of "good" numbers (having gcd>1 with all terms) is closed under taking multiples; the primitive minimal good numbers correspond to minimal transversals.

- Conjecture: There are only finitely many primitive good numbers; they involve only primes from the first few terms. The sequence is the set of all multiples of these primitive numbers, hence periodic.

Current best

We are constructing a rigorous proof based on the following outline:

1. Let $P(x)$ be the set of prime divisors of x. The condition $\gcd(a_{n+1}, a_i) > 1$ for all $i \le n$ means $P(a_{n+1}) \cap P(a_i) \neq \emptyset$ for all $i \le n$.
2. Define the family $\mathcal{F} = \{P(a_n) : n \ge 1\}$. A number $x > 1$ is "good" iff $P(x)$ is a transversal of $\mathcal{F}$ (i.e., $P(x) \cap E \neq \emptyset$ for every $E \in \mathcal{F}$).
3. The sequence (a_n) consists exactly of all good numbers $\ge a_1$ in increasing order. (Because each a_n is good, and if some good number x were missing, the greedy algorithm would have picked it when the sequence passed below it.)
4. Let $\mathcal{T}$ be the family of minimal transversals of $\mathcal{F}$ (with respect to inclusion). Then a number is good iff its prime set contains some member of $\mathcal{T}$.
5. Therefore the good numbers are the union over $H \in \mathcal{T}$ of the multiples of $\prod_{p \in H} p$. If $\mathcal{T}$ is finite, this is a finite union of arithmetic progressions, and the sequence is periodic.
6. **Key lemma**: $\mathcal{T}$ is finite. Moreover, all members of $\mathcal{T}$ consist of primes that appear in the first three terms of the sequence.
   - *Proof*: Let p be the smallest prime divisor of a_1. Then $a_2 = a_1 + p$, and $P(a_1) \cap P(a_2) = \{p\}$.
   - If all terms are divisible by p, the sequence is arithmetic ($T = 1$, $L = p$).
   - Otherwise, let $k \ge 3$ be the first index with $p \nmid a_k$. Then $P(a_k)$ contains a minimal transversal H of $\{P(a_1), \dots, P(a_{k-1})\}$ with $p \notin H$.
   - After adding $P(a_k)$, the singleton $\{p\}$ is no longer a transversal. The new minimal transversals are subsets of $\bigcup_{i=1}^{k} P(a_i)$.
   - One shows that the family $\mathcal{T}$ stabilizes within at most one more step, and that no prime first appearing after a_k can enter any minimal transversal (because such a prime would be redundant—the new term already contains an old minimal transversal hitting all earlier edges).
   - Hence $\mathcal{T}$ is a finite family of subsets of the finite set $\bigcup_{i=1}^{k+1} P(a_i)$.
7. With $\mathcal{T}$ finite, let $M = \operatorname{lcm}\{\prod_{p \in H} p : H \in \mathcal{T}\}$. Then the set of good numbers is a union of residue classes modulo M. Let T be the number of good residues modulo M that are $\ge a_1$ in each block? Actually, since the good numbers are exactly the numbers $\ge a_1$ that are $\equiv r \pmod{M}$ for some residue r in a fixed set R, the sorted list satisfies $a_{n+T} = a_n + M$, where T is the number of good residues in a complete system modulo M. (We need to check that the density is constant from the start; this holds if a_1 itself is the smallest good number, which is true because all smaller numbers are either ≤ 1 or coprime to a_1? Actually numbers between 1 and a_1 might be good; but the sequence starts at a_1, and the periodicity is checked from $n = 1$. We can adjust T and L accordingly.)

We still need to complete the details of step 6 (stabilization of minimal transversals) and step 7 (deriving exact periodicity from the start).

Full proof

(To be written after completing the rigorous arguments.)